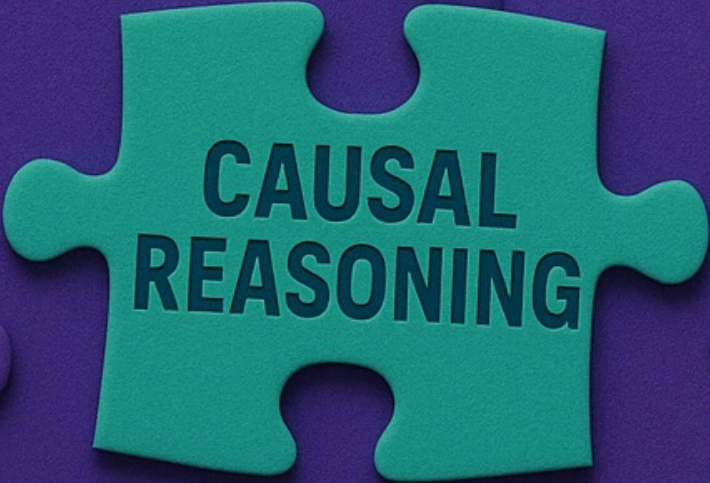


Part 3



**CAUSAL
REASONING**

THE PLAN!

- ❑ Indeterministic Causation
- ❑ Exercises

Indeterministic Causation

- In module 1 & 2, the causal relationships we studied were “deterministic,” that is, the effect is always completely determined by the values assigned to the causes.

Indeterministic Causation

- In module 1 & 2, the causal relationships we studied were “deterministic,” that is, the effect is always completely determined by the values assigned to the causes.
- *In an “indeterministic” causal relationship, the effect is not always determined by the causal assignments.*
- Almost every causal relationship we study in the social, behavioral, and medical sciences at least appears to be indeterministic.
 - Having wealthy parents is a cause of wealth, but it does not guarantee it. Eating well and exercising regularly cause good health, but they don’t guarantee it.

Indeterministic Causation

- Therefore, to capture real world phenomenon's, we cannot be satisfied with an account of causation among variables in which causes must completely determine the value of their effects.

Indeterministic Causation

- There are *three core ideas* in this module No 3 – ‘*Indeterministic Causation*’.
 1. The first is that causal relations are sometimes indeterministic.
 2. The second is that sometimes they are indeterministic in appearance only. In most cases we don't explicitly mention all of the causes that act on an effect of interest, but only some of them. If we did list all of the causes of a particular effect, then the relationship between the causes and the effect might well be deterministic. *When we leave some out, however, the relationship will appear to be indeterministic.*
 3. The third core idea is this: even though a particular causal assignment might not always determine whether a given effect will or will not occur, *it can determine the chances, or the probability that the effect will occur.*

Indeterministic Causation

- At the very least, switching from one causal assignment to another can change the chances that the effect will occur.
 - *Example - being exposed to asbestos, for example, does not guarantee that you will get cancer, but it does raise the chances that you will get it compared with not being exposed.*

The idea that causes only determine the probability of their effects is the main reason why causal science needs statistics.

Indeterministic Causation

- In what follows, we will look into 6 concepts:
 1. Indeterministic Response Structures
 2. Sources of Indeterminism
 3. Pseudo-indeterminism vs. True Indeterminism
 4. Quantifying Indeterministic Causation
 5. Definition of Indeterministic Causation
 6. Direct vs. Indirect Causation

Indeterministic Causation

(1) Indeterministic Response Structure

- **Deterministic Response Structure:** In a *deterministic response structure*, every causal assignment completely determines the value of the effect.
- **Indeterministic Response Structure:** In an *indeterministic response structure*, there is at least one causal assignment that does not determine the value of the effect.
 - Example: Cell Phones

Indeterministic Causation

- (2) Sources of Indeterminism
- **Response Structure Uniformity** - a population has response structure uniformity for a given effect if every individual in the population has the same response structure for that effect.

Notice that this definition doesn't say anything about whether each individual has a deterministic or indeterministic response structure, it only insists that they all have the *same* response structure.

Indeterministic Causation

- (2) Sources of Indeterminism
- **Response Structure Uniformity** - a population has response structure uniformity for a given effect if every individual in the population has the same response structure for that effect.
- **Indeterministic Causation in a Population** – if there are two individuals in a population who exhibit different effects even though they have the same causal assignment, then there is *apparent indeterministic causation in the population*. Causation in a population can appear indeterministic for two reasons:
 1. Some individual in the population has an indeterministic response structure.
 2. Every individual in the population has a deterministic response structure, but the population is not response structure uniform.

Indeterministic Causation

- Pseudo-indeterminism vs. True Indeterminism
 - **Pseudo-indeterministic Response Structure** - A response structure for an effect E is pseudo-indeterministic when i) it is an indeterministic response structure for E, and ii) there are other causes of E that, when added, make the response structure deterministic.
 - **Truly Indeterministic Response Structure** - A response structure for an effect E is truly-indeterministic when i) it is an indeterministic response structure for E, and ii) there is no set of causes we can add that will make it a deterministic response structure.

Indeterministic Causation

- (4) Quantifying Indeterminism?
 - We quantify indeterminism with probabilities, or chances.

RESPONSE STRUCTURE FOR SMOKING AND LUNG CANCER

Causal Assignment	Cause	Effect	
	SMOKER	LUNG CANCER = Yes	LUNG CANCER = No
1	Yes	15%	85%
2	No	1%	99%

- We list for each causal assignment the chances that the effect will be cancer and the chances that the effect will not be cancer.
- We don't know with certainty what the effect will be in any causal assignment, but using a table like this, we can express how the chances of the effect change as we change the causal assignment.

Indeterministic Causation

- (5) Definition of Indeterministic Causation
- In module 1 & 2
 - Test Pair of Causal Assignments for X Two causal assignments C1 and C2 are a *test pair of causal assignments* for X if and only if they are identical except for the values assigned to variable X
 - Direct Cause In a system of variables S, X is a direct cause of Y relative to S if and only if there is at least one test pair of causal assignments for X in the response structure for Y across which Y differs.
- The idea behind test pairs of causal assignments is to keep everything in the system the same except for one variable, which we make different between one causal assignment and another. If the effect changes, then we have causation.

Indeterministic Causation

- (5) Definition of Indeterministic Causation
 - What is it for the effect to change, however? Does it mean that the effect must take on a different value in the two causal assignments? Or is it enough that the chances for the effect change?
 - Clearly, in the case of indeterministic causation, only the chances need to change. So let's modify the definition of cause ever so slightly to accommodate indeterminism:
 - **Direct (Indeterministic) Cause In a system of variables S** , X is a direct cause of Y relative to S if and only if there is at least one test pair of causal assignments for X in the response structure Y across which the probability distribution over Y differs.
 - *The “probability distribution” means, roughly, the probability given to each of the possible values Y can take on. If the probability given to any of Y 's values changes, then the probability distribution over Y has changed.*

Indeterministic Causation

- (6) Direct vs. Indirect Causation
 - If X is a direct cause of Y relative to some causal system S , then we can hold the other variables in S constant and produce a change in Y as a result of changing X .
 - If X is only an indirect cause of Y , however, then all of X 's influence on Y must go through some other variable in S , say V . X must first produce a change in V , which must in turn produce a change Y .
 - In the response structure of Y , however, any test pair of X will involve two assignments in which V does not change, and so X will not appear as a direct cause of Y . That does *not* mean that X is not a cause of Y , it just means that it is not a *direct* cause of Y .

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

A - Reflect and discuss – A Concrete Example: Cell Phones - Cell Phone 1

Question No 01 – Did dialing the phone *always* result in the phone ringing? (choose one)

- A. Yes
- B. No

RESPONSE STRUCTURE FOR THE CELL PHONE SYSTEM	
Causal Assignment	Effect: Call Connected
BUTTON PUSHED = Send	???
BUTTON PUSHED = End	No

CELL PHONE SYSTEM	
Variable	Values
BUTTON PUSHED	[Send, End]
CALL CONNECTED	[Yes, No]

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

A - Reflect and discuss – A Concrete Example: Cell Phones - Cell Phone 1

Question No 01 – Did dialing the phone *always* result in the phone ringing? (choose one)

A. Yes

B. No

RESPONSE STRUCTURE FOR THE CELL PHONE SYSTEM

Causal Assignment	Effect: Call Connected
BUTTON PUSHED = Send	???
BUTTON PUSHED = End	No

CELL PHONE SYSTEM

Variable	Values
BUTTON PUSHED	[Send, End]
CALL CONNECTED	[Yes, No]

Hitting the End button does determine the effect: the call is not connected. The problem is assigning a value to the effect for the first causal assignment, BUTTON PUSHED = Send. We don't know what will happen to the effect in this causal assignment - we say that **this response structure is indeterministic**.

We separate deterministic from indeterministic causal relationships by separating deterministic vs. indeterministic response structures.

- **Deterministic Response Structure** - in a *deterministic response structure*, every causal assignment completely determines the value of the effect.
- **Indeterministic Response Structure** - in an *indeterministic response structure*, **there is at least one causal assignment** that does not determine the value of the effect.

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

A - Reflect and discuss – A Concrete Example: Cell Phones - Cell Phone 1

 Question No 02 – Which of the choices best represents the causal system in the simulation? (choose one)

- A. {Phone Button [send, end]}
- B. {Phone Button [send, end], Attempts[0,10]}
- C. {Phone Button [send, end], Call Connected [yes, no]}
- D. None of the above.

The simulated cell phone system, as we have presented it to you, includes two variables, each of which has two values:

CELL PHONE SYSTEM	
Variable	Values
BUTTON PUSHED	[Send, End]
CALL CONNECTED	[Yes, No]

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

A - Reflect and discuss – A Concrete Example: Cell Phones - Cell Phone 1

Question No 02 – Which of the choices best represents the causal system in the simulation? (choose one)

- A. {Phone Button [send, end]}
- B. {Phone Button [send, end], Attempts[0,10]}
- C. {Phone Button [send, end], Call Connected [yes, no]}**
- D. None of the above.

Answer C.

There are two variables in the system, the cause we can assign: Phone Button [send, end], and the effect we observe: Call Connected [yes, no].

The simulated cell phone system, as we have presented it to you, includes two variables, each of which has two values:

CELL PHONE SYSTEM	
Variable	Values
BUTTON PUSHED	[Send, End]
CALL CONNECTED	[Yes, No]

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

B - Reflect and discuss – A Concrete Example: Cell Phones - Cell Phone 2

 **Question No 03** – Consider the following response structure:

Causal Assignment	Exercises Regularly	Smokes	Impotent = yes
1	Yes	Yes	8%
2	Yes	No	2%
3	No	Yes	10%
4	No	No	2%

This response structure is indeterministic because: (choose one)

- A. It doesn't contain enough rows
- B. Exercise and Smoking have no influence on Impotence
- C. There are no test pairs for Exercise or Smoking
- D. There is at least one causal assignment that does not determine the value of the effect

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

B - Reflect and discuss – A Concrete Example: Cell Phones - Cell Phone 2

 **Question No 03** – Consider the following response structure:

Causal Assignment	Exercises Regularly	Smokes	Impotent = yes
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4	No	No	2%

This response structure is indeterministic because: (choose one)

- A. It doesn't contain enough rows
- B. Exercise and Smoking have no influence on Impotence
- C. There are no test pairs for Exercise or Smoking
- D. There is at least one causal assignment that does not determine the value of the effect**

Answer - D.

The response structure is indeterministic because there is at least one causal assignment in which the effect is not determined - in fact in this case there are 4.

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

B - Reflect and discuss – A Concrete Example: Cell Phones - Cell Phone 2

 **Question No 04 – The difference between a deterministic and an indeterministic response structure is that: (choose one)**

- A. There are no direct causes in an indeterministic response structure, while there are direct causes in deterministic response structures.
- B. In an indeterministic response structure, there is at least one causal assignment that does not determine the value of the effect.
- C. Indeterministic response structures always have fewer variables than deterministic ones.
- D. Deterministic response structures are based on evidence, while indeterministic response are matters of speculation.

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

B - Reflect and discuss – A Concrete Example: Cell Phones - Cell Phone 2

 Question No 04 – The difference between a deterministic and an indeterministic response structure is that: (choose one)

- A. There are no direct causes in an indeterministic response structure, while there are direct causes in deterministic response structures.
- B. In an indeterministic response structure, there is at least one causal assignment that does not determine the value of the effect.**
- C. Indeterministic response structures always have fewer variables than deterministic ones.
- D. Deterministic response structures are based on evidence, while indeterministic response are matters of speculation.

Answer - B.

An indeterministic response structure can have as many variables as you like. The correct answer is B.

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

C - Reflect and discuss - An Abstract Example: Red and Blue Squares – Red Squares I

 Question No 05 – Does making the state of the red square hatched fully determine the state of the blue square? (choose one)

- A. Yes
- B. No
- C. Neither

RESPONSE STRUCTURE FOR THE COLORED SQUARE EXPERIMENT

Causal Assignment	Effect: Blue Square
Red Square = Solid	???
Red Square = Hatched	Hatched

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

C - Reflect and discuss - An Abstract Example: Red and Blue Squares – Red Squares I

Question No 05 – Does making the state of the red square hatched fully determine the state of the blue square? (choose one)

- A. Yes
- B. No
- C. Neither

RESPONSE STRUCTURE FOR THE COLORED SQUARE EXPERIMENT

Causal Assignment	Effect: Blue Square
Red Square = Solid	???
Red Square = Hatched	Hatched

Answer - A.

By setting the red square to hatched, we guarantee that the blue square will be hatched. This is represented in the simulation by the fact that in the row for the 'Red Square = Hatched' assignment, the 'Effect = Solid' blue square will have 0 in it, and the other column, 'Effect = Hatched' blue square will have as many as the total, meaning that every trial in that assignment came out hatched.

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

C - Reflect and discuss - An Abstract Example: Red and Blue Squares – Red Squares I

 Question No 06 – Does making the state of the red square solid fully determine the state of the blue square? (choose one)

A. Yes

B. No

RESPONSE STRUCTURE FOR THE COLORED SQUARE EXPERIMENT

Causal Assignment	Effect: Blue Square
Red Square = Solid	???
Red Square = Hatched	Hatched

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

C - Reflect and discuss - An Abstract Example: Red and Blue Squares – Red Squares I

Question No 06 – Does making the state of the red square solid fully determine the state of the blue square? (choose one)

A. Yes

B. No

RESPONSE STRUCTURE FOR THE COLORED SQUARE EXPERIMENT	
Causal Assignment	Effect: Blue Square
Red Square = Solid	???
Red Square = Hatched	Hatched

Answer - B.

By setting the red square to solid, we do not guarantee that the blue square will be solid or hatched. This is represented in the simulation by the fact that in the row for the 'Red Square = Solid' assignment, the 'Effect = Solid' blue square will not have a 0 in it, and the other column, 'Effect = Hatched' blue square will not have 0 either, meaning that some trials resulted in one effect and some in other, even though the causal assignment was the same.

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

D - Reflect and discuss - Individuals and Populations 1

 Question No 07 – If, in a given population, we assume that every individual's response structure is deterministic, can we tell if a population violates response structure uniformity? (choose one)

- A. Yes, always
- B. No, never
- C. Yes, but only in certain circumstances

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

D - Reflect and discuss - Individuals and Populations 1

 **Question No 07 – If, in a given population, we assume that every individual's response structure is deterministic, can we tell if a population violates response structure uniformity? (choose one)**

- A. Yes, always
- B. No, never
- C. Yes, but only in certain circumstances**

Answer - C.

If everyone in the population has a deterministic response structure, we can sometimes tell if response structure uniformity is violated. For example, suppose individuals 3 and 5 differed in their response to a causal assignment, and they were both in that assignment. Since the response structure is deterministic, we could observe that they had the same causal assignment and different effects, and then conclude that uniformity must be violated. Even if everyone in the population has a deterministic response structure, we can't always tell if response structure uniformity holds.

- For example, suppose every individual was the same except for individual 25. If there are 4 different causal assignments, and individual 25 was different in the response to causal assignment 4, then uniformity is violated. If, however, individual 25 was the same as everyone else on causal assignments 1 thru 3, and individual 25 happened to be in assignment 1, we could not tell.

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

E - Reflect and discuss - Individuals and Populations 2

Question No 08 – Does the causal assignment that you give to the red square have the same effect on every member of the population ? (choose one)

- A. Yes
- B. No

RESPONSE STRUCTURE FOR THE COLORED SQUARE EXPERIMENT

Causal Assignment	Effect: Blue Square
Red Square = Solid	???
Red Square = Hatched	Hatched

In the Colored Square example above, you performed the same experiment over and over on the same pair of red and blue squares.

Suppose you are given a population of 20 distinct individual pairs. For each of the individual pairs, you can click on the red square to change its causal assignment. You can change all the red squares once, and notice how many of the blue squares match. Then you repeat the experiment, and again notice how many of the blue squares change.

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

E - Reflect and discuss - Individuals and Populations 2

Question No 08 – Does the causal assignment that you give to the red square have the same effect on every member of the population ?
(choose one)

- A. Yes
- B. No**

RESPONSE STRUCTURE FOR THE COLORED SQUARE EXPERIMENT

Causal Assignment	Effect: Blue Square
Red Square = Solid	???
Red Square = Hatched	Hatched

In the Colored Square example above, you performed the same experiment over and over on the same pair of red and blue squares.

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Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

E - Reflect and discuss - Individuals and Populations 2

Question No 09 – In the population in the simulation: (choose one)

- A. The response structure for the blue square is deterministic.
- B. The response structure for the blue square is indeterministic.
- C. None of the above.

RESPONSE STRUCTURE FOR THE COLORED SQUARE EXPERIMENT

Causal Assignment	Effect: Blue Square
Red Square = Solid	???
Red Square = Hatched	Hatched

In the Colored Square example above, you performed the same experiment over and over on the same pair of red and blue squares.

Suppose you are given a population of 20 distinct individual pairs. For each of the individual pairs, you can click on the red square to change its causal assignment. You can change all the red squares once, and notice how many of the blue squares match. Then you repeat the experiment, and again notice how many of the blue squares change.

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

E - Reflect and discuss - Individuals and Populations 2

Question No 09 – In the population in the simulation: (choose one)

- A. The response structure for the blue square is deterministic.
- B. The response structure for the blue square is indeterministic.**
- C. None of the above.

Answer - B.

There were almost certainly some pairs of red and blue squares that you gave the same causal assignment but that had different responses for the effect, thus the response structure is indeterministic.

RESPONSE STRUCTURE FOR THE COLORED SQUARE EXPERIMENT

Causal Assignment	Effect: Blue Square
Red Square = Solid	???
Red Square = Hatched	Hatched

In the Colored Square example above, you performed the same experiment over and over on the same pair of red and blue squares.

Suppose you are given a population of 20 distinct individual pairs. For each of the individual pairs, you can click on the red square to change its causal assignment. You can change all the red squares once, and notice how many of the blue squares match. Then you repeat the experiment, and again notice how many of the blue squares change.

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

F - Reflect and discuss – Cell Phone Revisited 1

*We only showed you part of the story in the cell phone example. Here we uncover another cause of getting the call to go through: **whether you are in range of a cell phone tower or not**. You cannot control whether you are in range of the tower – but you may be able to observe it.*

 **Question No 10 – In which of the following causal assignments results does the call go through? (choose one)**

- A. Situation 1: Call placed, and within tower range
- B. Situation 2: Call placed, and not within tower range
- C. Situation 3: Call not placed, and within tower range
- D. Situation 4: Call not placed, and not within tower range

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

F - Reflect and discuss – Cell Phone Revisited 1

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- C. Situation 3: Call not placed, and within tower range
- D. Situation 4: Call not placed, and not within tower range

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

F - Reflect and discuss – Cell Phone Revisited 1

Question No 11 – Is the response structure for “the call gets through” deterministic? (choose one)

A. Yes

B. No

RESPONSE STRUCTURE FOR THE FULL CELL PHONE SYSTEM

Causal Assignment	BUTTON PUSHED	IN TOWER RANGE	Effect: CALL CONNECTED
1	Send	Yes	Yes
2	Send	No	No
3	End	Yes	No
4	End	No	No

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

F - Reflect and discuss – Cell Phone Revisited 1

Question No 11 – Is the response structure for “the call gets through” deterministic? (choose one)

A. Yes

B. No

RESPONSE STRUCTURE FOR THE FULL CELL PHONE SYSTEM			
Causal Assignment	BUTTON PUSHED	IN TOWER RANGE	Effect: CALL CONNECTED
1	Send	Yes	Yes
2	Send	No	No
3	End	Yes	No
4	End	No	No

Answer - A.

If you assign values to “call placed” (yes or no) and “within tower range” (yes or no), you know with certainty the value of the effect variable.

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

G - Reflect and discuss – Red and Blue Squares Revisited 1

What happened in the cell phones is the same as what happened in the red and blue square simulation: we only gave you part of the story. In the following example you get the full picture.

Now, you can intervene to change the state of the red square, but the computer has control over a newly added green square. The causal assignment for the states of the red and green square is created partly by you clicking on the red square and partly by the computer choosing the state of the green square. When you change the state of the red square, the program assigns a value at random to the green square. Sometimes it assigns it hatched green and sometimes solid green. The red and green squares are both causes of the blue square.

Question No 12 – Should the missing cell in the Response Structure for The System of Squares contain “solid”, or “hatched”? (choose one)

- A. Solid
- B. Hatched

RESPONSE STRUCTURE FOR THE SYSTEM OF SQUARES			
Causal Assignment	Causal Factor 1: RED	Causal Factor 2: GREEN	Effect: BLUE
1	Solid	Solid	Solid
2	Solid	Hatched	Hatched
3	Hatched	Solid	
4	Hatched	Hatched	Hatched

RESPONSE STRUCTURE FOR THE COLORED SQUARE EXPERIMENT

Causal Assignment	Effect: Blue Square
Red Square = Solid	???
Red Square = Hatched	Hatched

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

G - Reflect and discuss – Red and Blue Squares Revisited 1

What happened in the cell phones is the same as what happened in the red and blue square simulation: we only gave you part of the story. In the following example you get the full picture.

Now, you can intervene to change the state of the red square, but the computer has control over a newly added green square. The causal assignment for the states of the red and green square is created partly by you clicking on the red square and partly by the computer choosing the state of the green square. When you change the state of the red square, the program assigns a value at random to the green square. Sometimes it assigns it hatched green and sometimes solid green. The red and green squares are both causes of the blue square.

Question No 12 – Should the missing cell in the Response Structure for The System of Squares contain “solid”, or “hatched”? (choose one)

- A. Solid
- B. Hatched**

Answer - B.

In all cases in which the causal assignment includes Red=Hatched, the Blue Square = Hatched.

RESPONSE STRUCTURE FOR THE SYSTEM OF SQUARES			
Causal Assignment	Causal Factor 1: RED	Causal Factor 2: GREEN	Effect: BLUE
1	Solid	Solid	Solid
2	Solid	Hatched	Hatched
3	Hatched	Solid	
4	Hatched	Hatched	Hatched

RESPONSE STRUCTURE FOR THE COLORED SQUARE EXPERIMENT

Causal Assignment	Effect: Blue Square
Red Square = Solid	???
Red Square = Hatched	Hatched

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

H - Reflect and discuss – Pseudo-Indeterminism 1

Question No 13 – Consider the response structure for malaria given in the table titled: Response Structure For Malaria – With Gene Unobserved. Suppose that we are told only that Joel was bitten by the mosquito and had no inoculation. Will he get malaria? (choose one)

- A. Yes
- B. No
- C. Impossible to tell

RESPONSE STRUCTURE FOR MALARIA – WITH GENE UNOBSERVED				
Causal Assignment	Causal Factor 1: BITTEN	Causal Factor 2: HAS GENE	Cusal Factor 3: INOCULATED	Effect: MALARIA
1	True	Unobserved	True	False
2	True	Unobserved	False	False
3	True	Unobserved	True	False
4	True	Unobserved	False	True
5	False	Unobserved	True	False
6	False	Unobserved	False	False
7	False	Unobserved	True	False
8	False	Unobserved	False	False

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

H - Reflect and discuss – Pseudo-Indeterminism 1

Question No 13 – Consider the response structure for malaria given in the table titled: Response Structure For Malaria – With Gene Unobserved. Suppose that we are told only that Joel was bitten by the mosquito and had no inoculation. Will he get malaria? (choose one)

- A. Yes
- B. No
- C. Impossible to tell**

RESPONSE STRUCTURE FOR MALARIA – WITH GENE UNOBSERVED				
Causal Assignment	Causal Factor 1: BITTEN	Causal Factor 2: HAS GENE	Cusal Factor 3: INOCULATED	Effect: MALARIA
1	True	Unobserved	True	False
2	True	Unobserved	False	False
3	True	Unobserved	True	False
4	True	Unobserved	False	True
5	False	Unobserved	True	False
6	False	Unobserved	False	False
7	False	Unobserved	True	False
8	False	Unobserved	False	False

Answer C.

Is it true that in all cases in which Joel was bitten and had no inoculation that Joel did not get malaria?

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

H - Reflect and discuss – Pseudo-Indeterminism 1

Question No 14 – Suppose that we are told that Joel was bitten by the mosquito but had an inoculation. **Which causal assignment might Joel have? (choose all that apply)**

- A. Assignment 1.
- B. Assignment 2.
- C. Assignment 3.
- D. Assignment 4.
- E. Assignment 5.
- F. Assignment 6.
- G. Assignment 7.
- H. Assignment 8.

RESPONSE STRUCTURE FOR MALARIA – WITH GENE UNOBSERVED

Causal Assignment	Causal Factor 1: BITTEN	Causal Factor 2: HAS GENE	Causal Factor 3: INOCULATED	Effect: MALARIA
1	True	Unobserved	True	False
2	True	Unobserved	False	False
3	True	Unobserved	True	False
4	True	Unobserved	False	True
5	False	Unobserved	True	False
6	False	Unobserved	False	False
7	False	Unobserved	True	False
8	False	Unobserved	False	False

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

H - Reflect and discuss – Pseudo-Indeterminism 1

Question No 14 – Suppose that we are told that Joel was bitten by the mosquito but had an inoculation. **Which causal assignment might Joel have? (choose all that apply)**

- A. Assignment 1.
- B. Assignment 2.
- C. Assignment 3.
- D. Assignment 4.
- E. Assignment 5.
- F. Assignment 6.
- G. Assignment 7.
- H. Assignment 8.

RESPONSE STRUCTURE FOR MALARIA – WITH GENE UNOBSERVED				
Causal Assignment	Causal Factor 1: BITTEN	Causal Factor 2: HAS GENE	Causal Factor 3: INOCULATED	Effect: MALARIA
1	True	Unobserved	True	False
2	True	Unobserved	False	False
3	True	Unobserved	True	False
4	True	Unobserved	False	True
5	False	Unobserved	True	False
6	False	Unobserved	False	False
7	False	Unobserved	True	False
8	False	Unobserved	False	False

Answer - A and C.

You need to find the rows in which Joel is assigned a “Yes” for the variable of being bitten and “Yes” for the variable of having an inoculation. There is more than one such row.

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

H - Reflect and discuss – Psuedo-Indeterminism 1

Question No 15 – If Joel was bitten by the mosquito but had an inoculation, **will he get malaria?** (choose one)

- A. Yes
- B. No
- C. Impossible to tell

RESPONSE STRUCTURE FOR MALARIA – WITH GENE UNOBSERVED

Causal Assignment	Causal Factor 1: BITTEN	Causal Factor 2: HAS GENE	Cusal Factor 3: INOCULATED	Effect: MALARIA
1	True	Unobserved	True	False
2	True	Unobserved	False	False
3	True	Unobserved	True	False
4	True	Unobserved	False	True
5	False	Unobserved	True	False
6	False	Unobserved	False	False
7	False	Unobserved	True	False
8	False	Unobserved	False	False

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

H - Reflect and discuss – Psuedo-Indeterminism 1

Question No 15 – If Joel was bitten by the mosquito but had an inoculation, **will he get malaria?** (choose one)

- A. Yes
- B. No**
- C. Impossible to tell

RESPONSE STRUCTURE FOR MALARIA – WITH GENE UNOBSERVED

Causal Assignment	Causal Factor 1: BITTEN	Causal Factor 2: HAS GENE	Cusal Factor 3: INOCULATED	Effect: MALARIA
1	True	Unobserved	True	False
2	True	Unobserved	False	False
3	True	Unobserved	True	False
4	True	Unobserved	False	True
5	False	Unobserved	True	False
6	False	Unobserved	False	False
7	False	Unobserved	True	False
8	False	Unobserved	False	False

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

I - Reflect and discuss – Cell Phones and Indeterminism 1

Question No 16 – Suppose that, instead of flipping a computer’s version of a fair coin, we programmed the simulation to roll a computer’s version of a regular six sided die - and when it rolled a “1” or a “2” we set IN TOWER RANGE to equal Yes, and when it rolled either a 3, 4, 5, or 6 we set IN TOWER RANGE to equal No. **What should we put into the box that now contains Yes = 50%, No = 50% in table: Response Structure for the Small Cell Phone System? (choose one)**

- A. Yes = 50%, No = 50%
- B. Yes = 33%, No = 67%
- C. Yes = 67%, No = 33%
- D. Yes = 16.67%, No = 83.33%

RESPONSE STRUCTURE FOR THE SMALL CELL PHONE SYSTEM		
Causal Assignment	BUTTON PUSHED	Effect: CALL CONNECTED
1	Send	Yes = 50%, No = 50%
2	End	No

In all the Cell Phone experiment trials, we could have a computer set the value of IN TOWER RANGE by flipping a computer’s version of a fair coin. About 50% of the time, IN TOWER RANGE should equal Yes, and about 50% of the time it should equal No. That means that about 50% of the time you pushed the Send button, the call went through, and 50% it didn’t. If you only did a few trials, you might have gotten a proportion quite different from 50% by luck, but if you did a lot of trials it’s very unlikely you got something very different from 50%. In any event, instead of putting “???” into the response structure for the small system, we could have put something more informative about our uncertainty, like: Yes = 50%, No = 50%.

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

I - Reflect and discuss – Cell Phones and Indeterminism 1

Question No 16 – Suppose that, instead of flipping a computer’s version of a fair coin, we programmed the simulation to roll a computer’s version of a regular six sided die - and when it rolled a “1” or a “2” we set IN TOWER RANGE to equal Yes, and when it rolled either a 3,4,5, or 6 we set IN TOWER RANGE to equal No. **What should we put into the box that now contains Yes = 50%, No = 50% in table: Response Structure for the Small Cell Phone System? (choose one)**

- A. Yes = 50%, No = 50%
- B. Yes = 33%, No = 67%**
- C. Yes = 67%, No = 33%
- D. Yes = 16.67%, No = 83.33%

RESPONSE STRUCTURE FOR THE SMALL CELL PHONE SYSTEM		
Causal Assignment	BUTTON PUSHED	Effect: CALL CONNECTED
1	Send	Yes = 50%, No = 50%
2	End	No

Answer - B.

Since the chance of the car being in range of the cell tower is now 33% instead of 50%, the chance of the call going through is also reduced from 50% to 33%, and the chance of the call not going through is raised from 50% to 67%.

In all the Cell Phone experiment trials, we could have a computer set the value of IN TOWER RANGE by flipping a computer’s version of a fair coin. About 50% of the time, IN TOWER RANGE should equal Yes, and about 50% of the time it should equal No. That means that about 50% of the time you pushed the Send button, the call went through, and 50% it didn’t. If you only did a few trials, you might have gotten a proportion quite different from 50% by luck, but if you did a lot of trials it’s very unlikely you got something very different from 50%. In any event, instead of putting “???” into the response structure for the small system, we could have put something more informative about our uncertainty, like: Yes = 50%, No = 50%.

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

J - Reflect and discuss – Quantifying Indeterministic Causation 1

Question No 17 – Suppose there were three different media reports about an experiment on the effect of TV on violence. All the newspapers reported that the researchers gave two different causal assignments to 200 seventh grade boys. For 100 of the boys, they allowed them to watch as much violent TV as they wanted over 3 months, and for the other 100, they allowed none. The effect they measured was whether the boys got into any fights during the 3 months of the experiment. The newspapers gave different descriptions of the results. **Which of the following quotes provides a fully quantified indeterministic response structure for the population the researchers analyzed? (choose all that apply)**

- A. “Researchers reported that the boys who were allowed to watch TV were twice as likely to get into a fight as those who were not”.
- B. “Researchers reported that 12% of the boys who were allowed to watch TV got into a fight, but only 6% of the boys who were not allowed to watch got into one”.
- C. “Researchers reported that the boys who were allowed to watch TV were significantly more likely to get into a fight than those who were not”.
- D. None of the above.

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

J - Reflect and discuss – Quantifying Indeterministic Causation 1

Question No 17 – Suppose there were three different media reports about an experiment on the effect of TV on violence. All the newspapers reported that the researchers gave two different causal assignments to 200 seventh grade boys. For 100 of the boys, they allowed them to watch as much violent TV as they wanted over 3 months, and for the other 100, they allowed none. The effect they measured was whether the boys got into any fights during the 3 months of the experiment. The newspapers gave different descriptions of the results. **Which of the following quotes provides a fully quantified indeterministic response structure for the population the researchers analyzed? (choose all that apply)**

- A. “Researchers reported that the boys who were allowed to watch TV were twice as likely to get into a fight as those who were not”.
- B. “Researchers reported that 12% of the boys who were allowed to watch TV got into a fight, but only 6% of the boys who were not allowed to watch got into one”.
- C. “Researchers reported that the boys who were allowed to watch TV were significantly more likely to get into a fight than those who were not”.
- D. None of the above.

Answer - B.

Saying that the boys who were allowed to watch TV were twice as likely to get into a fight as those who were not is not enough information to construct the probability distribution over the effect :Got in a Fight = yes, Got in a Fight = no, for either of the two causal assignments. It might be that the chances of getting in a fight for the TV exposed was 90%, and 45% for the unexposed, or it might be that the chances were 50% for the TV exposed and 25% for the unexposed, etc.

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

K - Reflect and discuss – Direct Causation with Indeterminism 1

Question No 18 –

Causal Assignment	Gender	Experience	Hired
1	Female	Yes	20%
2	Female	No	10%
3	Male	Yes	20%
4	Male	No	10%

In this response structure for Hired, which are the direct indeterministic causes of Hired? (choose one)

- A. Gender
- B. Experience
- C. Neither Gender nor Experience
- D. Both Gender and Experience

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

K - Reflect and discuss – Direct Causation with Indeterminism 1

Question No 18 –

Causal Assignment	Gender	Experience	Hired
1	Female	Yes	20%
2	Female	No	10%
3	Male	Yes	20%
4	Male	No	10%

In this response structure for Hired, which are the direct indeterministic causes of Hired? (choose one)

- A. Gender
- B. Experience**
- C. Neither Gender nor Experience
- D. Both Gender and Experience

Answer - B.

Gender is not an indeterministic cause because the probability of being hired in this tables is the same across the test pairs for gender: 1 and 3, 2 and 4.

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

K - Reflect and discuss – Direct Causation with Indeterminism 1

Question No 19 –

Causal Assignment	Flu Shot	Avg. Sleep Per Night	Avoided Flu
1	Yes	Over 8 hours	90%
2	Yes	Under 8 hours	50%
3	No	Over 8 hours	20%
4	No	Under 8 hours	10%

What test pairs are sufficient to determine that having a flu shot is a direct cause of avoiding the flu in this system? (choose all that apply)

- A. 1 and 2.
- B. 1 and 4.
- C. 1 and 3.
- D. 2 and 4.

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

K - Reflect and discuss – Direct Causation with Indeterminism 1

Question No 19 –

Causal Assignment	Flu Shot	Avg. Sleep Per Night	Avoided Flu
1	Yes	Over 8 hours	90%
2	Yes	Under 8 hours	50%
3	No	Over 8 hours	20%
4	No	Under 8 hours	10%

What test pairs are sufficient to determine that having a flu shot is a direct cause of avoiding the flu in this system? (choose all that apply)

- A. 1 and 2.
- B. 1 and 4.
- C. 1 and 3.
- D. 2 and 4.

Answer - C and D.

1 and 2 do not form a test pair for flu shot. To test for a direct cause you need to consider a test pair in which alleged cause differs in value.

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

K - Reflect and discuss – Direct Causation with Indeterminism 1

Question No 20 –

Causal Assignment	Drinks Red Wine	Drinks Coffee	Eats Red Meat	Heart Disease
1	Yes	Yes	Yes	50%
2	Yes	Yes	No	60%
3	Yes	No	Yes	50%
4	Yes	No	No	60%
5	No	Yes	Yes	70%
6	No	Yes	No	50%
7	No	No	Yes	70%
8	No	No	No	50%

Which of the following are direct indeterministic causes of Heart Disease, according to this indeterministic response structure? (choose all that apply)

- A. Drinks red wine.
- B. Drinks coffee.
- C. Eats red meat.
- D. None of the above.

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

K - Reflect and discuss – Direct Causation with Indeterminism 1

Question No 20 –

Causal Assignment	Drinks Red Wine	Drinks Coffee	Eats Red Meat	Heart Disease
1	Yes	Yes	Yes	50%
2	Yes	Yes	No	60%
3	Yes	No	Yes	50%
4	Yes	No	No	60%
5	No	Yes	Yes	70%
6	No	Yes	No	50%
7	No	No	Yes	70%
8	No	No	No	50%

Which of the following are direct indeterministic causes of Heart Disease, according to this indeterministic response structure? (choose all that apply)

- A. Drinks red wine.
- B. Drinks coffee.
- C. Eats red meat.
- D. None of the above.

Answer - A and C.

The test pairs for Drinks Coffee are assignments 1 and 3, 2 and 4, 5 and 7, and 6 and 8. The probability of Heart Disease is the same across all of these test pairs.

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

K - Reflect and discuss – Direct Causation with Indeterminism 1

 Question No 21 –

Causal Assignment	Recitation Attendance	Studying Time	Got an A
1	Occasionally	High	10%
2	Moderately	High	42%
3	Frequently	High	10%
4	Occasionally	Low	10%
5	Moderately	Low	42%
6	Frequently	Low	10%

In this case, what are the test pairs for Studying Time? (choose one)

- A. 1 and 4, 2 and 5, 3 and 6
- B. 1 and 2, 2 and 3
- C. 1 and 2, 2 and 3, 1 and 3
- D. 4 and 5, 5 and 6, 4 and 6

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

K - Reflect and discuss – Direct Causation with Indeterminism 1

Question No 21 –

Causal Assignment	Recitation Attendance	Studying Time	Got an A
1	Occasionally	High	10%
2	Moderately	High	42%
3	Frequently	High	10%
4	Occasionally	Low	10%
5	Moderately	Low	42%
6	Frequently	Low	10%

In this case, what are the test pairs for Studying Time? (choose one)

- A. 1 and 4, 2 and 5, 3 and 6
- B. 1 and 2, 2 and 3
- C. 1 and 2, 2 and 3, 1 and 3
- D. 4 and 5, 5 and 6, 4 and 6

Answer - A.

To test for a direct cause you need to consider a test pair in which alleged cause differs in value. Studying Time is not different across assignments 1 and 2.

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

K - Reflect and discuss – Direct Causation with Indeterminism 1

Question No 22 –

Causal Assignment	Recitation Attendance	Studying Time	Got an A
1	Occasionally	High	10%
2	Moderately	High	42%
3	Frequently	High	10%
4	Occasionally	Low	10%
5	Moderately	Low	42%
6	Frequently	Low	10%

In this response structure, which are the direct indeterministic causes? (choose all that apply)

- A. Recitation Attendance.
- B. Study Time.
- C. Neither Recitation nor Study Time.

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

K - Reflect and discuss – Direct Causation with Indeterminism 1

Question No 22 –

Causal Assignment	Recitation Attendance	Studying Time	Got an A
1	Occasionally	High	10%
2	Moderately	High	42%
3	Frequently	High	10%
4	Occasionally	Low	10%
5	Moderately	Low	42%
6	Frequently	Low	10%

In this response structure, which are the direct indeterministic causes? (choose all that apply)

- A. Recitation Attendance.
- B. Study Time.
- C. Neither Recitation nor Study Time.

Answer - A.

Check all the relevant test pairs (there are three pairs) where Recitation Attendance has the same value and Study Time differs. Is there any difference in the probability of the effect?

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

L - Reflect and discuss Direct vs. Indirect Causation 1

Question No 23 – Consider the following response structure for the set of variables {Speeds Excessively [yes, no], Traffic Infraction [yes, no], Lose License [yes, no]}, where Lose License is treated as the effect.

Causal Assignment	Speeds Excessively	Traffic Infraction	Lose License = yes
1	Yes	Yes	65%
2	Yes	No	1%
3	No	Yes	65%
4	No	No	1%

This response structure tells us that: (choose one)

- A. Speeds Excessively is a direct cause of Lose License
- B. Speeds Excessively is not a direct cause of Lose License
- C. Speeds Excessively is an indirect cause of Lose License
- D. Speeds Excessively is not an indirect cause of Lose License

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

L - Reflect and discuss Direct vs. Indirect Causation 1

Question No 23 – Consider the following response structure for the set of variables {Speeds Excessively [yes, no], Traffic Infraction [yes, no], Lose License [yes, no]}, where Lose License is treated as the effect.

Causal Assignment	Speeds Excessively	Traffic Infraction	Lose License = yes
1	Yes	Yes	65%
2	Yes	No	1%
3	No	Yes	65%
4	No	No	1%

This response structure tells us that: (choose one)

- A. Speeds Excessively is a direct cause of Lose License
- B. Speeds Excessively is not a direct cause of Lose License**
- C. Speeds Excessively is an indirect cause of Lose License
- D. Speeds Excessively is not an indirect cause of Lose License

Answer - B.

The test pairs for Speeds Excessively are 1 and 3, and 2 and 4. The probability distribution over Lose License is the same across each of these test pairs, so Speeds Excessively is not a direct cause of Loses License.

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

L - Reflect and discuss Direct vs. Indirect Causation 1

Question No 24 – Consider the set of variables {Parents' Education, Intelligence, Education, Income}. Suppose you want to know whether Education is a cause of Income, either direct or indirect. To answer this question, you need to examine the response structure that:

- A. Contains the variables Intelligence, Education, and Income, where Income is treated as the effect.
- B. Contains only the variables Education and Income, where Income is treated as the effect.
- C. Contains only the variables Intelligence and Income, where Income is treated as the effect.
- D. Contains only the variables Education and Income, where Education is treated as the effect.

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

L - Reflect and discuss Direct vs. Indirect Causation 1

Question No 24 – Consider the set of variables {Parents' Education, Intelligence, Education, Income}. Suppose you want to know whether Education is a cause of Income, either direct or indirect. To answer this question, you need to examine the response structure that:

- A. Contains the variables Intelligence, Education, and Income, where Income is treated as the effect.
- B. Contains only the variables Education and Income, where Income is treated as the effect.**
- C. Contains only the variables Intelligence and Income, where Income is treated as the effect.
- D. Contains only the variables Education and Income, where Education is treated as the effect.

Answer - B.

Since we want to know whether Education is a cause of Income, we need to examine a response structure that includes both of these variables. A is incorrect because that response structure will only tell us whether or not Education is a direct cause of Income relative to Parents' Education, Intelligence, Income. But what we wanted to know was whether it was a cause, either direct or indirect.

Reading Exercise – No 15 (Module 3 | Indeterministic Causation)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

M - Reflect and discuss – Link it to your research proposal



Many of the ideas you have learned in this handout will be relevant to your Semester Research Project. Discuss the connection of these ideas to your project here in as much detail as possible, seeking help from your instructor when needed. (Refer back to Barroga and Matanguihan (2022) for more help. If your project has not progressed to this stage, you should come back to this activity later.)